

Unlight RL AI MVP: Technical Specification & PRD

Optimized for Automated Execution via GitHub Codespace AI Agent (Lean CPU-Only Edition)

Target Framework: Python 3.10+ / Gymnasium	Target Agent: Copilot Workspace / Cursor Agent
Algorithm: Masked PPO (sb3-contrib)	Compute Constraint: CPU 2-Core / 8GB RAM (Codespace)
Game Mode: 1v1 Single Character Turn-based	Interface: Rich Terminal CLI + FastAPI Endpoint

1. System Prompt for AI Agent (Copy-Paste Ready)

SYSTEM DIRECTIVE FOR CODESPACE AGENT:
You are acting as an expert Reinforcement Learning (RL) and Game Engineer. Your task is to implement a lean, fully functional MVP of a turn-based 1v1 card duel game inspired by Unlight, complete with a Gymnasium environment, Action Masked PPO Agent, and a playable Terminal CLI interface.

STRICT CONSTRAINTS:

- Write clean, modular, and type-annotated Python 3.10+ code.
- Keep CPU computation minimal so tests run in < 30 seconds inside GitHub Codespace.
- Follow the project repository structure outlined in Section 2 precisely.

2. Project Directory Structure & Blueprint

```
unlight_rl_mvp/  
  README.md           # Project overview and one-line run commands  
  requirements.txt     # Dependency list (gymnasium, sb3-contrib, torch, rich, fastapi)  
  main.py             # Entry point: CLI Human vs AI play & Fast-train mode  
  config.py           # Environment hyperparameter definitions  
  engine/  
    __init__.py  
    card.py           # Card definitions (Move, Sword, Gun, Special/Def)  
    character.py       # Character stats & skills (Ebert: Lightning/Precision Shot)  
    duel.py           # Turn state machine & dice roll resolver  
  env/  
    __init__.py  
    unlight_gym_env.py # Gymnasium custom environment implementation  
    action_mask.py     # Action space mapping & valid action mask calculator  
  agents/  
    __init__.py  
    rule_agent.py      # Heuristic Rule-based Bot (Baseline benchmark)  
    ppo_agent.py       # Masked PPO Model trainer wrapper (SB3-Contrib)  
  ui/  
    __init__.py  
    cli.py            # Terminal interactive interface (Rich library formatted)
```

3. Unlight Core Rules & MDP Formulation

Simplified MVP Game Rules (Ebert vs Ebert 1v1):

- **Distance System:** Range from 1 (Near) to 3 (Far). Default starting distance = 2.
- **Deck & Hand:** 4 Card types: Move (M), Sword (S), Gun (G), Special (Sp). Max Hand Size = 6.
- **Turn Phase Flow:** Draw Phase -> Move Phase (Bidding Move cards) -> Attack Phase (Sword at R=1, Gun at R=2-3) -> Defense Phase -> End Phase.
- **Ex Skills:** Ebert Skill 1 (Thunderbolt: Special 1+ Gun 2+), Ebert Skill 2 (Precision Shot: Gun 3+ Sword 1+).

Observation Space (1D Vector Representation):

Feature Vector Slice	Dim	Range	Description
Self / Opponent HP	2	[0, 10]	Normalized health values (Initial HP = 10)
Current Distance	1	[1, 3]	Current spatial range between players
Current Game Phase	1	[0, 3]	0: Move, 1: Attack, 2: Defense, 3: Round End
Self Hand Cards Count	4	[0, 6]	Counts for Card Types: [Move, Sword, Gun, Special]
Opponent Hand Count	1	[0, 6]	Total cards held by opponent (Partial Information)

Action Space & Masking Strategy:

To enable ultra-fast CPU training, the multi-card choice is mapped to a **Discrete Action Space (Size = 16)** representing strategic intentions (e.g., 0: Pass/Do Nothing, 1: Max Move Forward, 2: Max Move Back, 3: Basic Attack, 4: Trigger Skill 1, 5: Trigger Skill 2, etc.). An **Action Mask Array (Bool [16])** dynamically invalidates actions when the required cards are missing from the player's hand.

4. Reward Shaping & Self-Play Pipeline

Event Trigger	Reward Value	Rationale / Objective
Win Match	+1.0	Terminal Sparse Reward for victory
Lose Match	-1.0	Terminal Penalty for defeat
Deal Damage	+0.05 * Damage	Dense reward to encourage aggressive card usage
Take Damage	-0.04 * Damage	Dense penalty to incentivize defensive card timing
Trigger Ex Skill	+0.10	Bonus reward for executing tactical card combinations

Training Roadmap for Agent Execution:

1. **Baseline Phase:** Train Masked PPO against RuleAgent for 20,000 steps (~2 mins on Codespace CPU). Ensure win rate exceeds 75%.
2. **Self-Play Phase:** Instantiate opponent wrapper using freeze weights of previous PPO checkpoints. Train for 30,000 steps to learn robust counter-strategies.

5. Step-by-Step Action Plan for AI Agent

Task ID	Target Module	Action Items & Acceptance Criteria
TASK-01	requirements.txt	Define pins: gymnasium, sb3-contrib, torch, rich, fastapi, uvicorn.
TASK-02	engine/	Build duel state machine, card deck shuffling, Ebert skills & dice roller.
TASK-03	env/	Implement UnlightGymEnv (Gymnasium API) with action_masks() method.
TASK-04	agents/	Build RuleAgent and MaskablePPO trainer script with evaluation loop.
TASK-05	ui/ & main.py	Construct Rich CLI interface allowing human keyboard interaction vs trained AI.